DO YOU KNOW?

The word "Porous" is used by the wood anatomists to refer to the appearance of the vessels as pores in transverse section.

Differences Between Diffuse Porous Wood and Ring Porous Wood

Diffuse porous wood	Ring porous wood
This type of wood is formed where the climatic conditions are uniform.	This type of wood is formed where the climatic conditions are not uniform.
The vessels are more or less equal in diameter in any annual ring.	The vessels are wide and narrow within any annual ring.
The vessels are uniformly distributed throughout the wood.	The vessels are not uniformly distributed throughout the wood.

Tyloses

In many dicot plants, the lumen of the xylem vessels is blocked by many balloon-like ingrowths from the neighbouring parenchymatous cells. These balloon-like structures are called **tyloses** (fig 10.9).

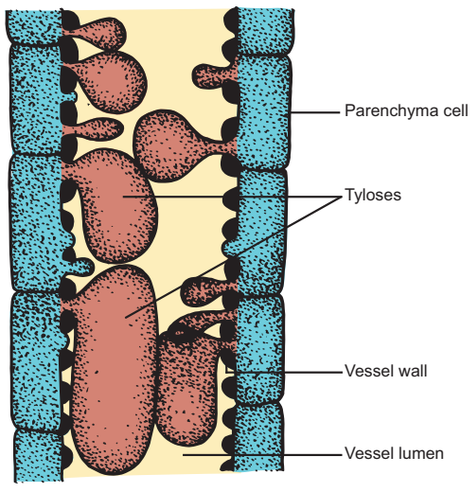


Figure 10.9: Structure of tyloses

Usually, these structures are formed in secondary xylem vessels that have last their function i.e., in heart wood.

In fully developed tyloses, starchy crystals, resins, gums, oils, tannins or coloured substances are found.

DO YOU KNOW?

There are tylosoids in gymnosperms and angiosperms

In **gymnosperms**, the resin ducts are blocked by tylose- like ingrowths from the neighbouring resin producing parenchyamatous cells. Example: *Pinus*.

In **angiosperms**, the sieve tubes are blocked by tylose- like ingrowths from the neighbouring parenchyamatous cells. Example: *Bombox*.

These are called **tylosoids**

Wood is also classified into **sap wood** and **heart wood**.

Sap Wood and Heart Wood

Sap wood and heart wood can be distinguished in the secondary xylem. In any tree the outer part of the wood, which is paler in colour, is called **sap wood or alburnum.** The centre part of the wood, which is darker in colour is called **heart wood or duramen.** The sap wood conducts water while the heart wood stops conducting water. As vessels of the heart wood are blocked by tyloses, water is not conducted through them. Due to the presence of tyloses and their contents the heartwood becomes coloured, dead and the hardest part of the wood.

From the economic point of view, generally the heartwood is more useful than the sapwood. The timber from the heartwood is more durable and more resistant to the attack of microorganisms and insects than the timber from sapwood.

DO YOU KNOW?

When, the heart wood of a tree is destroyed, no vital function of the plant is affected.

When, the sap wood is destroyed, the plant will die because conduction of water will be blocked.

11th Botany_EM_Chapter 10.indd 223 05-01-2023 18:28:33